

Share Base Stability and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Share Base Stability (SBS), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on SBS achieves an annualized gross (net) Sharpe ratio of 0.53 (0.47), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 23 (22) bps/month with a t-statistic of 2.90 (2.89), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (1 year), Growth in book equity, Net Payout Yield, Share issuance (5 year), Change in equity to assets, Momentum and LT Reversal) is 15 bps/month with a t-statistic of 2.00.

1 Introduction

Financial markets continuously process vast amounts of information, yet the speed and completeness of this incorporation remains a central question in asset pricing. While the efficient market hypothesis suggests that prices should rapidly reflect all available information, mounting evidence documents systematic patterns in returns that challenge this view. We identify a novel predictor of stock returns based on share base stability—the degree to which a firm’s outstanding shares remain constant over time—that generates economically significant return predictability. This finding suggests that markets systematically underreact to information contained in changes to firms’ equity capital structures, with important implications for our understanding of price formation and market efficiency.

The slow diffusion of information provides a compelling economic mechanism linking share base stability to future returns. When firms alter their share count through issuances, repurchases, or other corporate actions, these changes contain valuable signals about management’s private information and future prospects

The gradual incorporation of information about share base changes should manifest as return continuation, particularly where informational frictions are most severe. Small firms with limited analyst coverage experience slower information diffusion due to reduced intermediation and lower investor scrutiny [Hong et al. \(2000\)](#). Similarly, firms with dispersed ownership structures and limited institutional holdings face greater coordination frictions that impede price discovery [Boehmer and Wu \(2013\)](#). These cross-sectional predictions align with broader evidence on post-earnings-announcement drift and momentum, where underreaction concentrates among neglected stocks

Share base stability captures the cumulative information content of multiple corporate actions that individually may receive insufficient attention. While investors might partially process major equity issuances, the aggregation of smaller

changes—including option exercises, warrant conversions, and employee stock grants—often escapes immediate notice [Hirshleifer et al. \(2009\)](#). This inattention problem becomes particularly acute when multiple information sources compete for processing capacity, as documented in studies of earnings announcements clustering and Friday news releases

Our empirical analysis reveals that share base stability strongly predicts cross-sectional returns. A value-weighted long-short strategy that buys stocks with high SBS and sells those with low SBS generates an average monthly return of 33 basis points with a t-statistic of 4.17, translating to an annualized Sharpe ratio of 0.53. The strategy’s performance remains robust after controlling for known risk factors, delivering a monthly alpha of 23 basis points (t-statistic = 2.90) relative to the Fama-French six-factor model. These results establish share base stability as an economically meaningful predictor that captures information beyond existing factor models.

The predictive power of share base stability persists across various portfolio construction methodologies and remains significant after accounting for transaction costs. Net of trading costs, the strategy achieves a monthly return of 29 basis points with a t-statistic of 3.68, while the net generalized alpha relative to the six-factor model equals 22 basis points (t-statistic = 2.89). Importantly, the strategy generates significant returns even among the largest stocks, earning 23 basis points per month (t-statistic = 2.55) within the highest NYSE size quintile, consistent with slow information diffusion affecting even liquid securities.

Share base stability demonstrates incremental predictive power beyond closely related anomalies in the factor zoo. When we control for the six most closely related strategies—including share issuance measures, growth in book equity, and net payout yield—the SBS strategy maintains a significant alpha of 15 basis points per month (t-statistic = 2.00). Furthermore, the strategy’s gross Sharpe ratio of 0.53

exceeds 92% of documented anomalies, while its net Sharpe ratio of 0.47 ranks in the 98th percentile, highlighting its exceptional risk-adjusted performance relative to the broader cross-section of return predictors.

This paper makes several contributions to the asset pricing literature on market efficiency and information diffusion. While [Pontiff and Woodgate \(2008\)](#) document that share issuance predicts returns and [Daniel and Titman \(2006\)](#) show that composite issuance measures capture mispricing, our share base stability measure provides a more comprehensive signal that aggregates multiple dimensions of equity capital changes. Unlike discrete issuance events studied in [Loughran and Ritter \(1995\)](#), our continuous stability metric captures the cumulative information content of gradual share base evolution, including often-overlooked changes from employee compensation and derivative conversions.

Our findings extend the literature on limited attention and gradual information diffusion in several important ways. Building on [Hong and Stein \(1999\)](#) and [Hong et al. \(2000\)](#), who demonstrate that information diffuses slowly across investors, we show that even readily observable changes in shares outstanding experience delayed price incorporation. This evidence complements [Cohen and Frazzini \(2008\)](#) on customer-supplier momentum and [Cohen and Lou \(2012\)](#) on complicated firms, suggesting that investors struggle to process information that requires aggregation across multiple sources or time periods. Our results particularly reinforce [Hirshleifer and Teoh \(2003\)](#) and [Hirshleifer et al. \(2009\)](#) on limited attention to public information.

Methodologically, we demonstrate that share base stability survives the comprehensive robustness tests prescribed by [Novy-Marx and Velikov \(2023\)](#), generating significant net returns after transaction costs and maintaining predictive power when controlling for over 200 documented anomalies. The strategy expands the mean-variance efficient frontier even for investors with access to established factor models

and related anomalies, suggesting genuine economic content beyond data mining concerns raised by [Harvey et al. \(2016\)](#). These findings have important implications for market efficiency, indicating that prices incompletely reflect information about corporate financing decisions, and for investment practice, identifying an implementable strategy that enhances risk-adjusted returns net of realistic trading costs.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to capture year-over-year changes in firms' share base characteristics. Share Base Stability signal relies on two key accounting variables from firms' balance sheets. Common stock (CSTK) represents the par or stated value of all common stock issued by the company, reflecting the nominal value of shares outstanding multiplied by their par value. Common equity (CEQ) measures the total common shareholders' equity, calculated as the sum of common stock, capital surplus, and retained earnings, less treasury stock. This variable represents the book value of equity attributable to common shareholders and serves as a natural scaling factor for changes in the share base. We construct the Share Base Stability signal by computing the year-over-year change in common stock scaled by lagged common equity: $(\text{CSTK}(t) - \text{CSTK}(t-1)) / \text{CEQ}(t-1)$. This ratio captures the relative change in a firm's share base, with positive values indicating share issuance activity and negative values suggesting share repurchases or retirements. The scaling by lagged common equity normalizes the change relative to the firm's equity base, making the measure comparable across firms of different sizes. We calculate this signal using fiscal year-end values and require non-missing

observations for both CSTK and CEQ in consecutive years. To ensure meaningful economic interpretation, we exclude observations where lagged CEQ is negative or zero, as these cases typically represent firms in financial distress where the scaling would produce unreliable metrics.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the SBS signal. Panel A plots the time-series of the mean, median, and interquartile range for SBS. On average, the cross-sectional mean (median) SBS is -0.01 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input SBS data. The signal’s interquartile range spans -0.01 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the SBS signal for the CRSP universe. On average, the SBS signal is available for 6.39% of CRSP names, which on average make up 7.69% of total market capitalization.

4 Does SBS predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on SBS using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high SBS portfolio and sells the low SBS portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short SBS strategy earns an average return of 0.33% per month with a t-statistic of 4.17. The annualized Sharpe

ratio of the strategy is 0.53. The alphas range from 0.23% to 0.35% per month and have t-statistics exceeding 2.90 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.30, with a t-statistic of 5.48 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 590 stocks and an average market capitalization of at least \$1,490 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 24 bps/month with a t-statistics of 3.24. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for eighteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the

generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 21-29bps/month. The lowest return, (21 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.75. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the SBS trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-five cases.

Table 3 provides direct tests for the role size plays in the SBS strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and SBS, as well as average returns and alphas for long/short trading SBS strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the SBS strategy achieves an average return of 23 bps/month with a t-statistic of 2.55. Among these large cap stocks, the alphas for the SBS strategy relative to the five most common factor models range from 17 to 24 bps/month with t-statistics between 1.85 and 2.59.

5 How does SBS perform relative to the zoo?

Figure 2 puts the performance of SBS in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio

histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the SBS strategy falls in the distribution. The SBS strategy’s gross (net) Sharpe ratio of 0.53 (0.47) is greater than 92% (98%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the SBS strategy (red line).² Ignoring trading costs, a \$1 invested in the SBS strategy would have yielded \$8.58 which ranks the SBS strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the SBS strategy would have yielded \$6.24 which ranks the SBS strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the SBS relative to those. Panel A shows that the SBS strategy gross alphas fall between the 66 and 73 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The SBS strategy has a positive net generalized alpha for five out of the five factor models. In these cases SBS ranks between the 82 and 89 percentiles in terms of how

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

much it could have expanded the achievable investment frontier.

6 Does SBS add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of SBS with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price SBS or at least to weaken the power SBS has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of SBS conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{SBS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{SBS}SBS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{SBS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on SBS. Stocks are finally grouped into

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

five SBS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted SBS trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on SBS and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the SBS signal in these Fama-MacBeth regressions exceed 0.85, with the minimum t-statistic occurring when controlling for Momentum and LT Reversal. Controlling for all six closely related anomalies, the t-statistic on SBS is -1.43.

Similarly, Table 5 reports results from spanning tests that regress returns to the SBS strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the SBS strategy earns alphas that range from 16-26bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.02, which is achieved when controlling for Momentum and LT Reversal. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the SBS trading strategy achieves an alpha of 15bps/month with a t-statistic of 2.00.

7 Does SBS add relative to the whole zoo?

Finally, we can ask how much adding SBS to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria

(blue lines) or these 154 anomalies augmented with the SBS signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes SBS grows to \$2810.42.

8 Conclusion

This study provides compelling evidence that Share Base Stability (SBS) represents a robust and economically significant predictor of cross-sectional equity returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that SBS-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for well-established risk factors and closely related anomalies.

The key findings of our research underscore the economic importance of SBS as an asset pricing signal. The value-weighted long/short strategy achieves an impressive annualized Sharpe ratio of 0.53 gross (0.47 net), indicating strong performance even after accounting for transaction costs. More importantly, the strategy’s ability to generate significant alpha relative to the Fama-French five-factor model plus momentum (23 bps/month gross with t-statistic of 2.90) suggests that SBS captures a distinct dimension of expected returns not explained by conventional risk factors. The persistence of a 15 bps monthly alpha (t-statistic of 2.00) even after controlling

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which SBS is available.

for six closely related strategies—including various share issuance measures, payout yield, and momentum effects—further validates SBS as an independent source of return predictability rather than a mere repackaging of existing anomalies.

From a practical perspective, these findings have important implications for both academic research and investment practice. For academics, SBS contributes to our understanding of how corporate financing decisions and share structure dynamics influence asset prices. For practitioners, the signal’s robust performance and reasonable Sharpe ratio after transaction costs suggest potential implementation value, particularly as part of a diversified quantitative strategy. The signal’s distinctiveness from related factors also implies potential diversification benefits when combined with other systematic strategies.

However, several limitations of this study warrant consideration. First, while we employ the Novy-Marx and Velikov (2023) protocol to ensure robustness, our analysis is necessarily constrained by the available sample period and may not fully capture the signal’s performance across all market regimes. Second, the implementation assumptions, while conservative, may not fully reflect the market impact and capacity constraints that large-scale investors might face. Third, our analysis focuses primarily on U.S. equities, leaving the international applicability of SBS unexplored.

Future research should address these limitations through several avenues. First, examining SBS in international markets would provide valuable out-of-sample validation and insights into the signal’s universality. Second, investigating the economic mechanisms underlying the SBS premium—whether driven by risk-based explanations, behavioral biases, or institutional frictions—would enhance our theoretical understanding. Third, exploring the interaction between SBS and corporate events such as MA activity, spin-offs, or changes in ownership structure could reveal additional nuances in the signal’s predictive power. Finally, examining the signal’s performance during different market conditions and its sensitivity to various im-

plementation choices would provide practitioners with more granular guidance for real-world application.

In conclusion, Share Base Stability emerges as a robust and economically meaningful predictor of equity returns that merits inclusion in the canon of documented asset pricing anomalies. While further research is needed to fully understand its economic drivers and practical boundaries, the evidence presented here establishes SBS as a valuable addition to both the academic literature and the practitioner's toolkit.

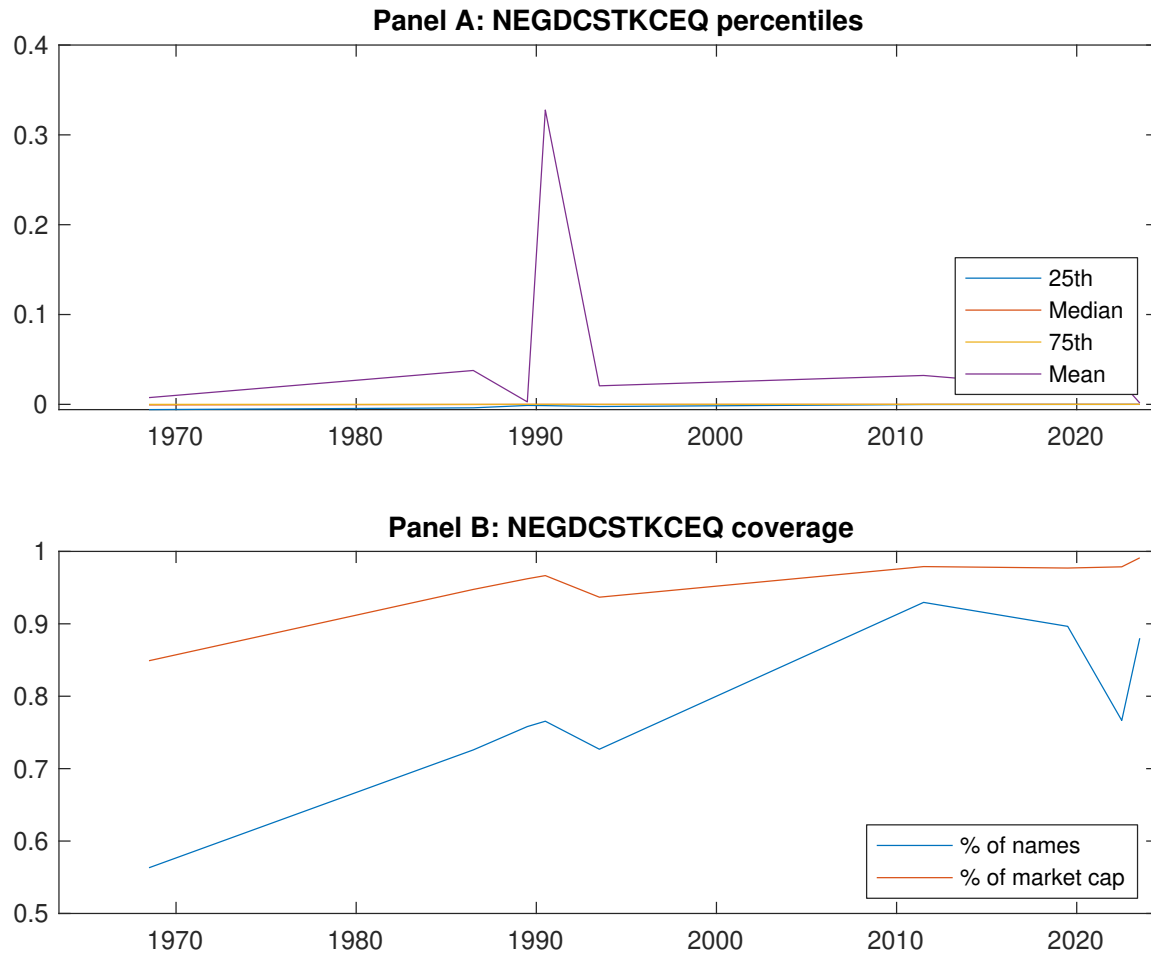


Figure 1: Times series of SBS percentiles and coverage.
This figure plots descriptive statistics for SBS. Panel A shows cross-sectional percentiles of SBS over the sample. Panel B plots the monthly coverage of SBS relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on SBS. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on SBS-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.45 [2.70]	0.49 [2.70]	0.64 [3.63]	0.64 [3.96]	0.78 [4.85]	0.33 [4.17]
α_{CAPM}	-0.12 [-2.20]	-0.14 [-3.08]	0.03 [0.74]	0.09 [1.87]	0.23 [4.94]	0.35 [4.41]
α_{FF3}	-0.13 [-2.39]	-0.12 [-2.80]	0.05 [1.19]	0.05 [1.06]	0.19 [4.17]	0.31 [3.98]
α_{FF4}	-0.10 [-1.92]	-0.09 [-2.03]	0.05 [1.18]	0.02 [0.52]	0.17 [3.73]	0.27 [3.42]
α_{FF5}	-0.15 [-2.81]	-0.07 [-1.60]	0.08 [1.68]	-0.04 [-0.91]	0.11 [2.49]	0.26 [3.27]
α_{FF6}	-0.13 [-2.41]	-0.05 [-1.06]	0.08 [1.65]	-0.05 [-1.16]	0.10 [2.31]	0.23 [2.90]
Panel B: Fama and French (2018) 6-factor model loadings for SBS-sorted portfolios						
β_{MKT}	0.97 [76.32]	1.03 [96.13]	1.02 [91.60]	1.00 [94.31]	0.98 [90.38]	0.01 [0.49]
β_{SMB}	-0.02 [-0.90]	0.02 [1.26]	0.04 [2.45]	-0.06 [-4.21]	0.00 [0.22]	0.02 [0.72]
β_{HML}	0.07 [3.05]	-0.02 [-0.89]	-0.07 [-3.34]	0.05 [2.50]	0.04 [2.12]	-0.03 [-0.83]
β_{RMW}	0.13 [5.29]	-0.09 [-4.29]	-0.07 [-3.30]	0.12 [5.99]	0.11 [5.41]	-0.02 [-0.45]
β_{CMA}	-0.11 [-3.04]	-0.08 [-2.80]	0.01 [0.30]	0.21 [7.11]	0.19 [6.15]	0.30 [5.48]
β_{UMD}	-0.03 [-2.22]	-0.03 [-3.17]	0.00 [0.05]	0.02 [1.59]	0.01 [0.89]	0.04 [1.97]
Panel C: Average number of firms (n) and market capitalization (me)						
n	721	693	590	683	747	
me (\$10 ⁶)	1764	1490	2140	2265	2475	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the SBS strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.33 [4.17]	0.35 [4.41]	0.31 [3.98]	0.27 [3.42]	0.26 [3.27]	0.23 [2.90]
Quintile	NYSE	EW	0.47 [8.45]	0.52 [9.47]	0.46 [8.78]	0.41 [7.76]	0.37 [7.19]	0.34 [6.48]
Quintile	Name	VW	0.32 [4.07]	0.34 [4.30]	0.30 [3.86]	0.27 [3.38]	0.25 [3.21]	0.23 [2.91]
Quintile	Cap	VW	0.24 [3.24]	0.26 [3.39]	0.24 [3.11]	0.20 [2.52]	0.21 [2.80]	0.18 [2.37]
Decile	NYSE	VW	0.33 [3.86]	0.33 [3.78]	0.28 [3.27]	0.24 [2.78]	0.27 [3.14]	0.24 [2.77]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.29 [3.68]	0.31 [4.00]	0.28 [3.57]	0.25 [3.27]	0.24 [3.10]	0.22 [2.89]
Quintile	NYSE	EW	0.27 [4.39]	0.33 [5.34]	0.27 [4.62]	0.25 [4.19]	0.18 [3.17]	0.17 [2.92]
Quintile	Name	VW	0.28 [3.58]	0.30 [3.84]	0.26 [3.41]	0.24 [3.15]	0.23 [2.98]	0.22 [2.81]
Quintile	Cap	VW	0.21 [2.75]	0.22 [2.90]	0.20 [2.60]	0.17 [2.29]	0.19 [2.47]	0.17 [2.25]
Decile	NYSE	VW	0.29 [3.35]	0.28 [3.28]	0.24 [2.82]	0.22 [2.57]	0.24 [2.77]	0.22 [2.58]

Table 3: Conditional sort on size and SBS

This table presents results for conditional double sorts on size and SBS. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on SBS. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high SBS and short stocks with low SBS. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	SBS Quintiles					SBS Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.47 [1.91]	0.68 [2.65]	0.87 [3.40]	0.90 [3.85]	0.96 [4.11]	0.52 [7.23]	0.54 [7.54]	0.51 [7.14]	0.47 [6.48]	0.44 [6.11]	0.41 [5.65]
	(2)	0.62 [2.72]	0.68 [2.87]	0.85 [3.62]	0.96 [4.26]	0.91 [4.22]	0.30 [3.55]	0.34 [4.01]	0.28 [3.39]	0.26 [3.05]	0.24 [2.88]	0.23 [2.67]
	(3)	0.55 [2.74]	0.66 [3.05]	0.74 [3.39]	0.83 [4.09]	0.95 [4.84]	0.40 [5.34]	0.41 [5.40]	0.38 [5.01]	0.37 [4.70]	0.36 [4.58]	0.35 [4.38]
	(4)	0.51 [2.74]	0.65 [3.26]	0.76 [3.77]	0.77 [4.13]	0.79 [4.32]	0.28 [3.81]	0.29 [3.99]	0.23 [3.28]	0.22 [3.12]	0.12 [1.72]	0.13 [1.78]
	(5)	0.48 [2.93]	0.47 [2.62]	0.52 [2.99]	0.51 [3.09]	0.71 [4.52]	0.23 [2.55]	0.24 [2.59]	0.22 [2.33]	0.17 [1.85]	0.21 [2.24]	0.18 [1.88]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	SBS Quintiles					SBS Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	383	384	384	382	374	31	34	40	29	29	
	(2)	109	107	108	108	107	57	56	58	57	55	
	(3)	78	78	78	77	78	99	96	98	100	100	
	(4)	66	65	66	66	66	206	204	215	213	217	
(5)	60	60	60	60	60	1479	1493	1661	1628	1887		

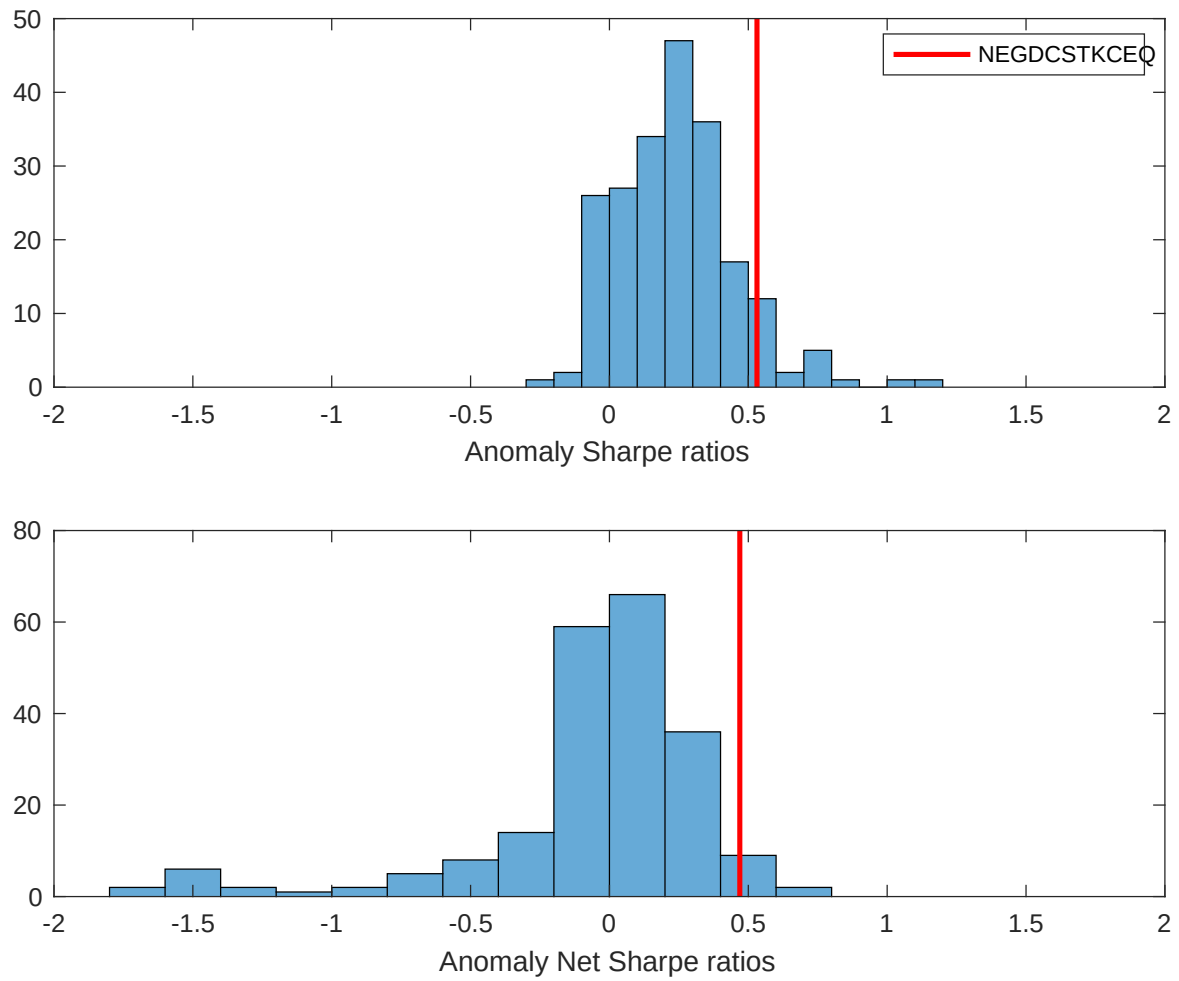


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the SBS with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

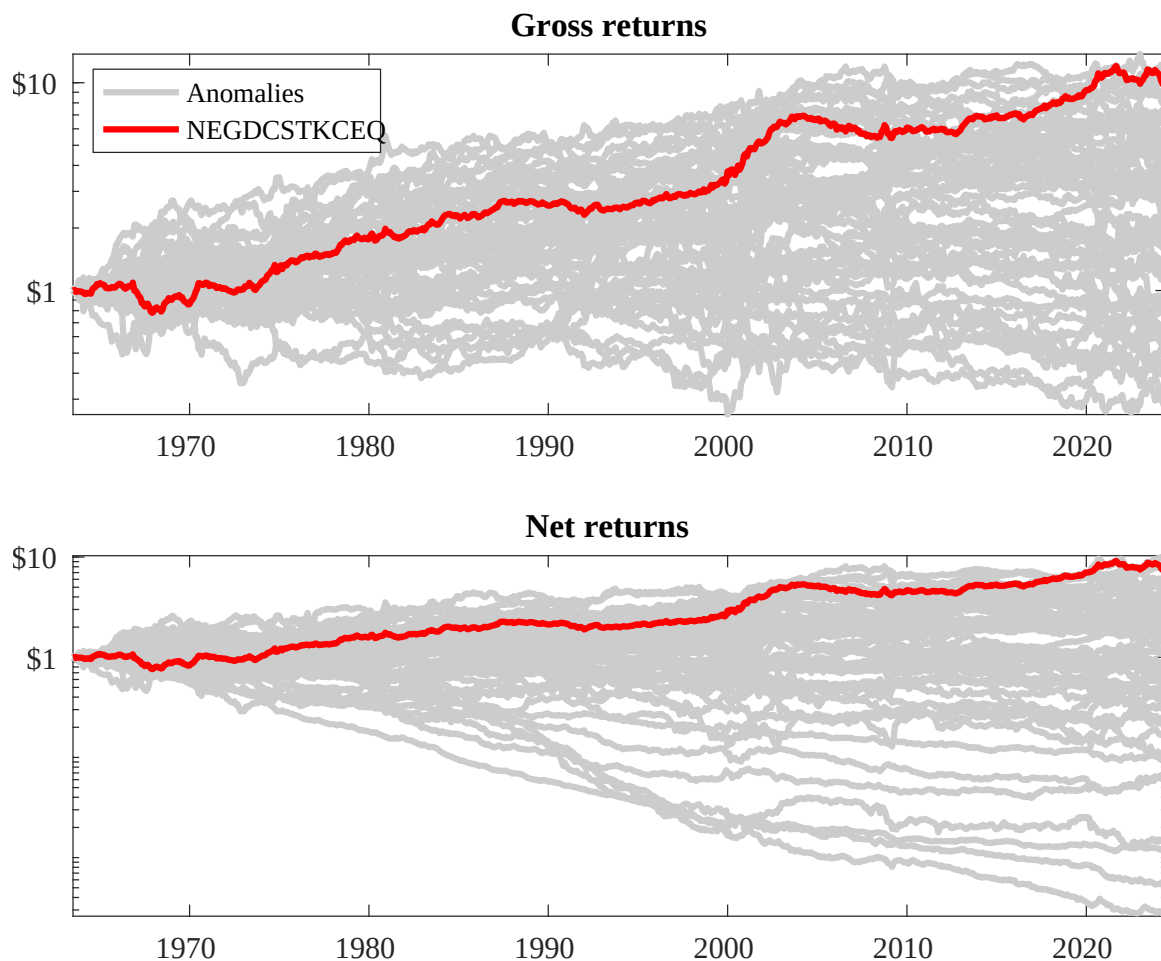
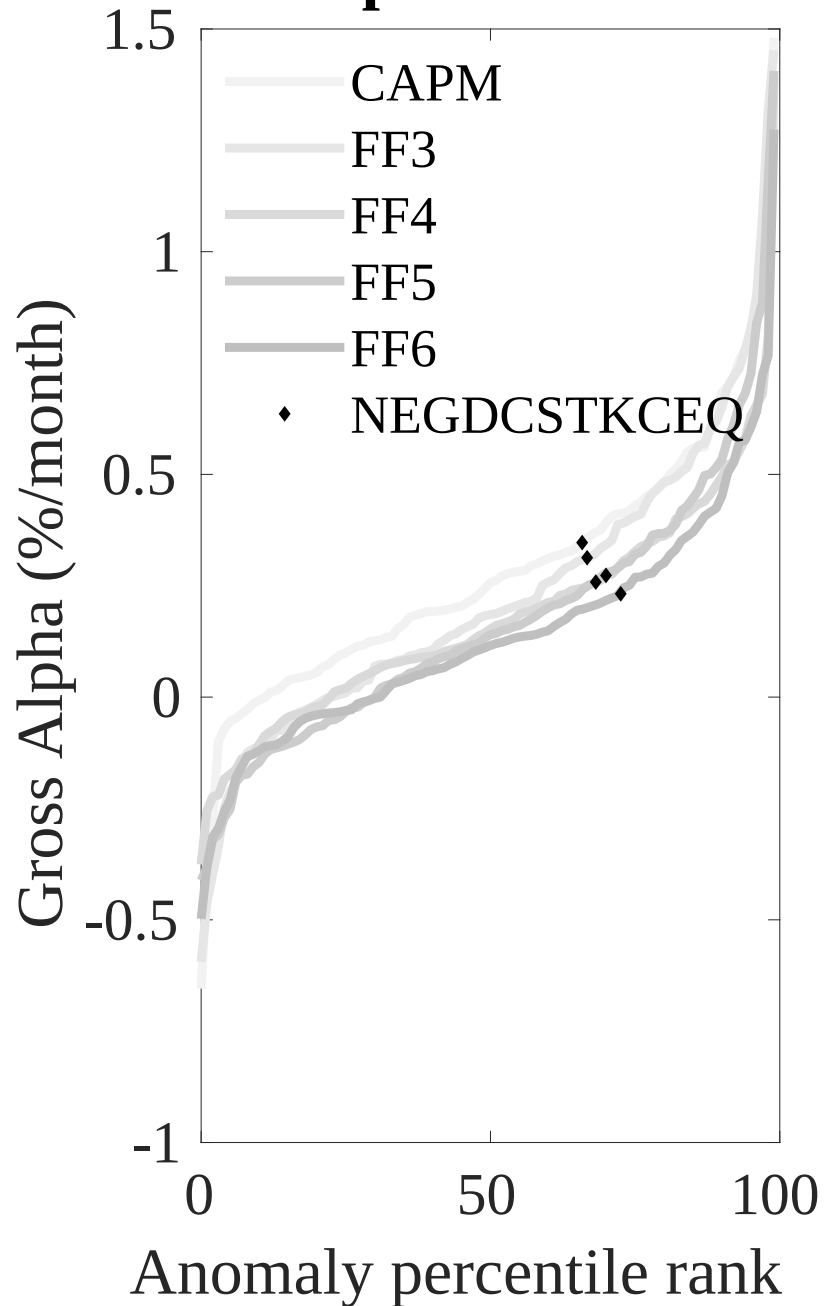


Figure 3: Dollar invested.

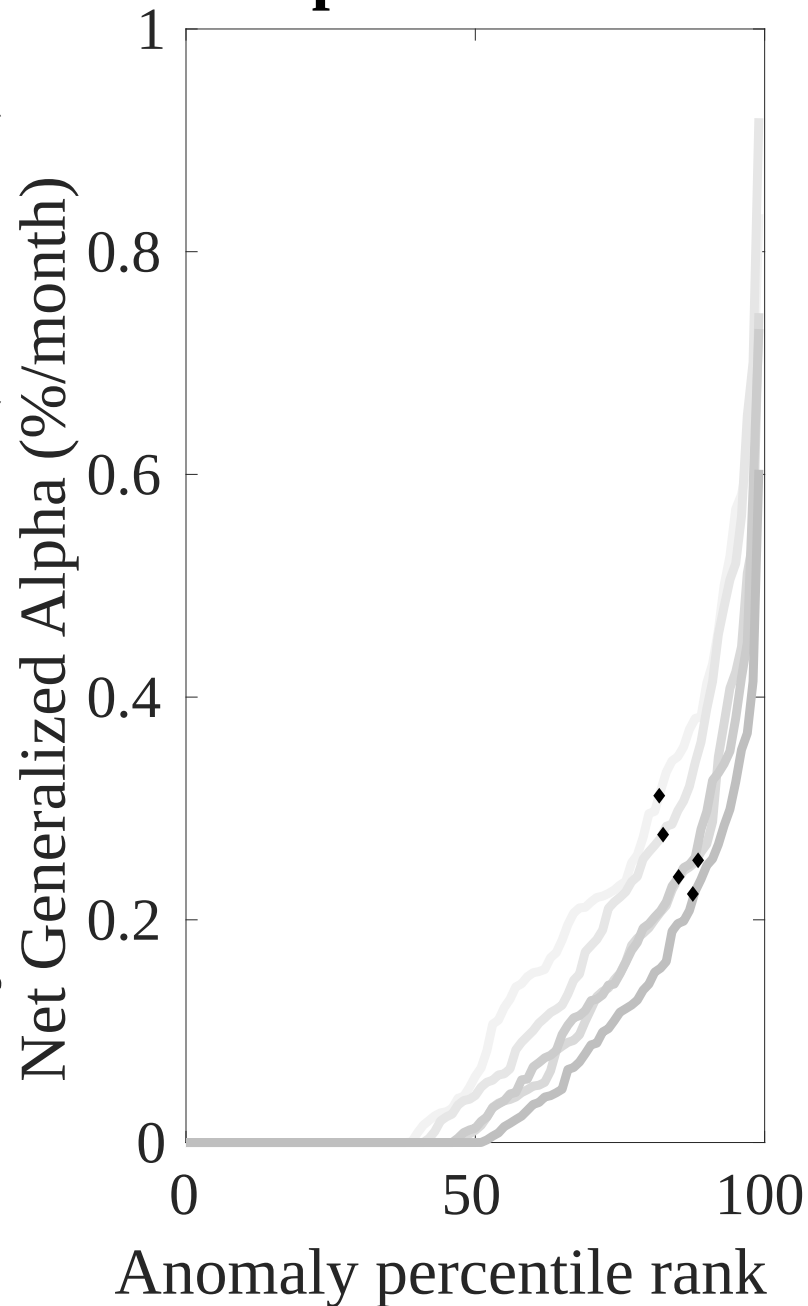
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the SBS trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



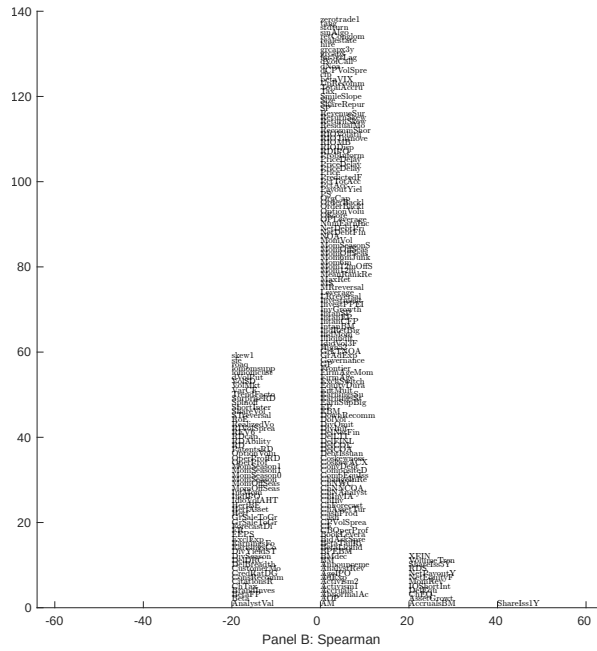
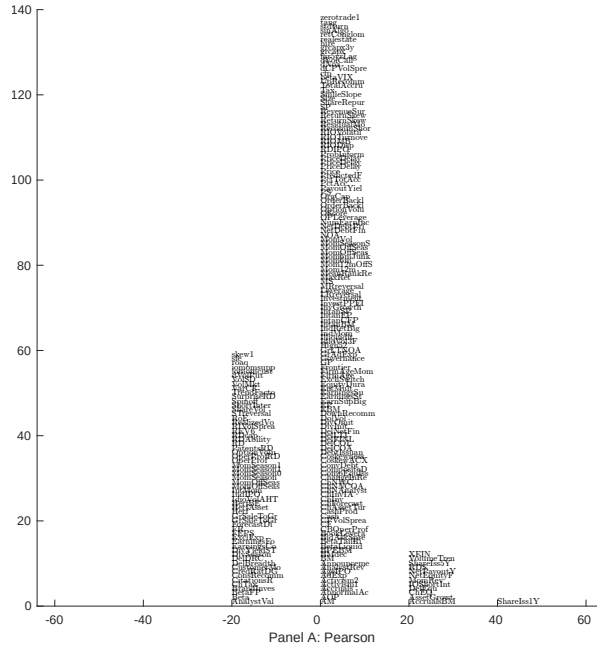


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with SBS. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

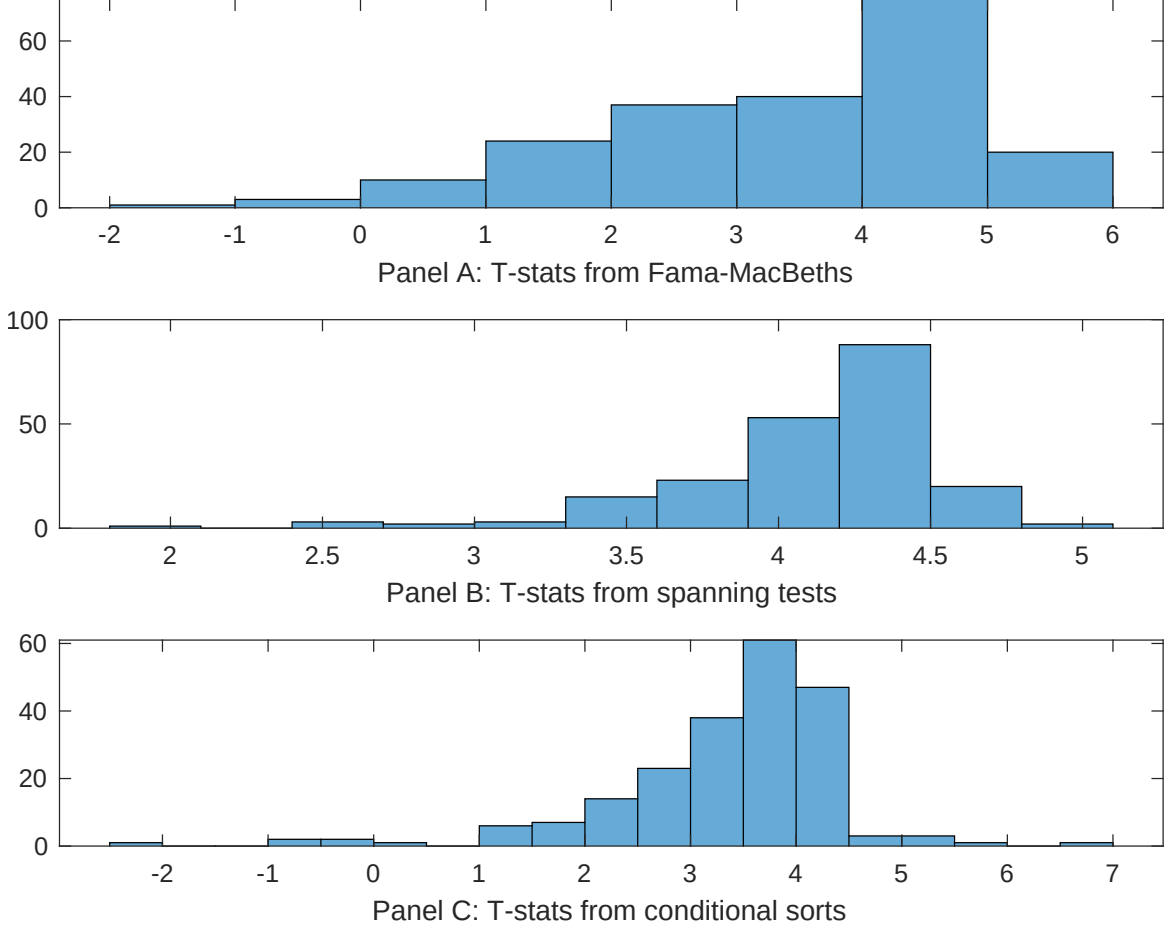


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of SBS conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{SBS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{SBS}SBS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{SBS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on SBS. Stocks are finally grouped into five SBS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted SBS trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on SBS. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{SBS}SBS_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Share issuance (1 year), Growth in book equity, Net Payout Yield, Share issuance (5 year), Change in equity to assets, Momentum and LT Reversal. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [6.04]	0.16 [6.83]	0.12 [5.43]	0.13 [6.43]	0.12 [5.68]	0.38 [1.12]	0.12 [2.73]
SBS	0.19 [3.45]	0.22 [3.64]	0.19 [2.68]	0.17 [3.07]	0.22 [3.72]	0.29 [0.85]	-0.42 [-1.43]
Anomaly 1	0.13 [5.75]						0.61 [1.21]
Anomaly 2		0.38 [3.18]					0.44 [1.83]
Anomaly 3			0.20 [1.80]				0.13 [1.48]
Anomaly 4				0.26 [4.71]			0.26 [2.32]
Anomaly 5					0.11 [3.08]		-0.13 [-1.56]
Anomaly 6						0.11 [4.27]	0.10 [3.48]
# months	715	720	715	715	720	648	510
$\bar{R}^2(\%)$	0	0	1	0	0	2	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the SBS trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{SBS} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Share issuance (1 year), Growth in book equity, Net Payout Yield, Share issuance (5 year), Change in equity to assets, Momentum and LT Reversal. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.21 [2.79]	0.22 [2.86]	0.26 [3.32]	0.16 [2.02]	0.23 [2.96]	0.21 [2.65]	0.15 [2.00]
Anomaly 1	30.47 [7.88]						14.56 [3.14]
Anomaly 2		31.31 [7.56]					31.55 [5.39]
Anomaly 3			15.84 [5.24]				5.20 [1.60]
Anomaly 4				30.15 [7.35]			15.58 [3.38]
Anomaly 5					11.63 [2.97]		-17.59 [-3.28]
Anomaly 6						5.09 [4.77]	4.19 [4.14]
mkt	1.86 [1.02]	2.14 [1.17]	3.58 [1.87]	2.47 [1.34]	0.90 [0.48]	1.28 [0.68]	4.37 [2.41]
smb	4.54 [1.72]	2.25 [0.85]	5.88 [2.14]	5.35 [2.01]	3.15 [1.16]	1.32 [0.49]	3.60 [1.38]
hml	-2.40 [-0.68]	-5.40 [-1.52]	-7.69 [-2.02]	0.83 [0.23]	-3.07 [-0.84]	-2.81 [-0.77]	-5.82 [-1.61]
rmw	-18.17 [-4.40]	-0.17 [-0.05]	-10.87 [-2.69]	-12.47 [-3.23]	-0.58 [-0.16]	0.96 [0.26]	-16.23 [-3.76]
cma	9.97 [1.76]	-1.52 [-0.23]	15.49 [2.65]	14.26 [2.57]	17.10 [2.57]	26.36 [4.96]	-6.64 [-1.01]
umd	2.59 [1.43]	3.42 [1.89]	5.24 [2.82]	2.01 [1.10]	4.27 [2.29]	-0.97 [-0.46]	-2.39 [-1.18]
# months	728	732	728	728	732	728	728
$\bar{R}^2(\%)$	14	13	10	13	8	9	22

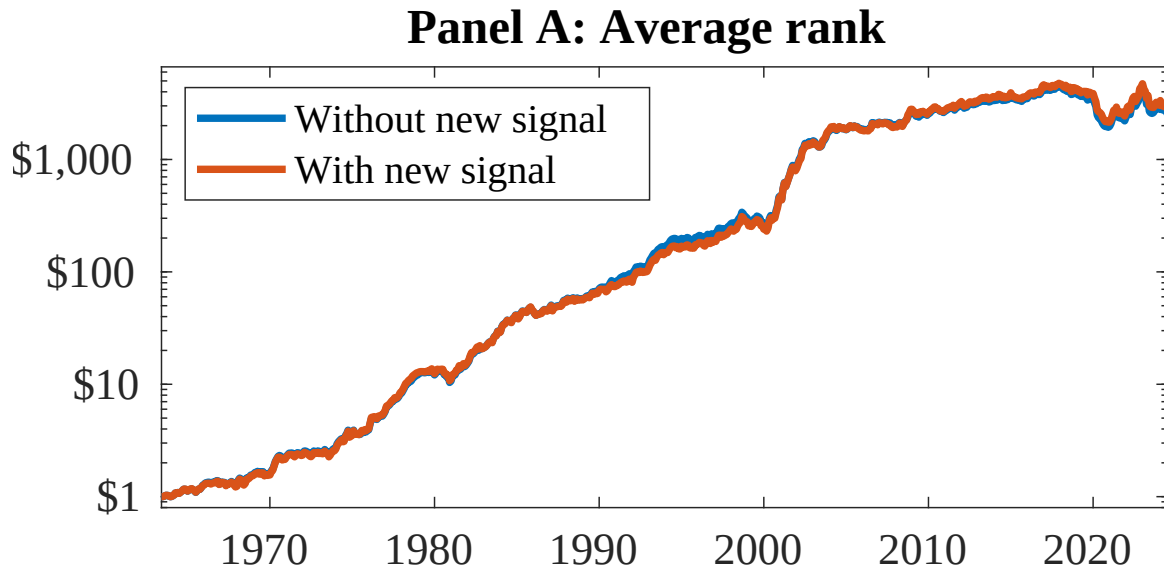


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as SBS. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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